Spatial Patterns in Child Food Hardship and School Nutrition Policy

Byeol Choi(byeolchoi98) from section 1
Takuma Kazama(tak1610) from section 1

1. Motivation

Child food hardship remains unevenly distributed across U.S. states.
Federal school nutrition programs — especially the Community Eligibility Provision (CEP) and the National School Lunch Program (NSLP) — aim to reduce this burden by expanding meal access.

This project asks three connected questions:

- Do higher CEP participation rates correspond to lower food hardship?
- Does higher NSLP spending per school-age child relate to improved outcomes?

Rather than focusing only on regression estimates, this project emphasizes **visual and spatial evidence**.

2. Data and Construction

State level visualization

For state-level visulaization, we mainly focus on child food insufficiency data (Househould Pulse Survey), CEP participation data (Food and Research Action Center), and 2024 Presidential election voting data (MIT Election Lab). For data construction, we created a master data (master_state_marged.csv), which also has other child well-bieng variables (details are in README.qmd) as well as the main variables. For the voting data, which is at the county level, we aggregated it at the state level.

Difficulties or spcial steps in the data cleaning process

- Constructing Food Insufficiency:
  Food insufficiency is defined as households reporting "sometimes" or "often" not having enough to eat in the past 7 days, weighted by total respondents.
- Democratic margin:
  The voting data only has the number of votes that each party gets, so we converted it into a relative metric, which is called "democratic margin".

County level visualization

For county-level visualization, we focus on the same variables of interest as the state-level visualization. However, food insecurity data is not available at the county level, so we used county-level child poverty rate (U.S. Census Bureau) as a proxy. For eahc data, we subset them to a county-level one.

Difficulties or spcial steps in the data cleaning process

- CEP participation rate: CEP participation rate data is the shcool-district level, so we used IL's ISBE code as a crosswalk, which connects school ID with county fips codes in IL.

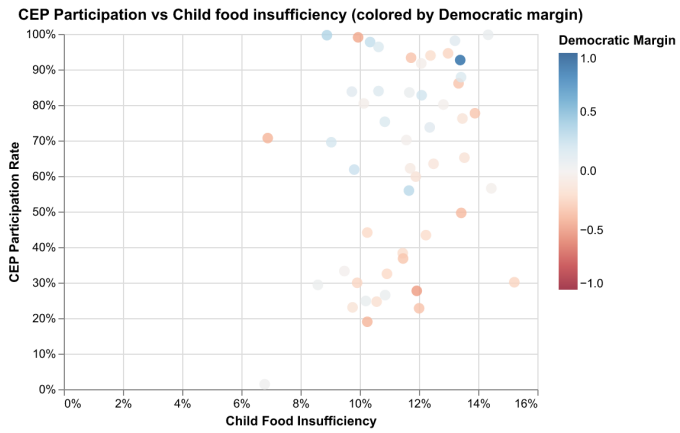
OLS regression analysis (state-level)

For OLS regression, we add additional variables to the master data for more robust analysis (master_merged_with_controls.csv). Detailed information is descrived in README.qmd.

3. Visualizing Geographic Variation

State level visualization

To address the question "Do higher CEP participation rates correspond to lower food hardship?", we plot child food insufficiency rate and cep participation rate data in a scatterplot. In addtion to this, we colored the points by democratic margin because our interest also lies in whether some other factor than child well-bieng variables may explain cep participation rate. To do so, we merged voting data and master data as well.



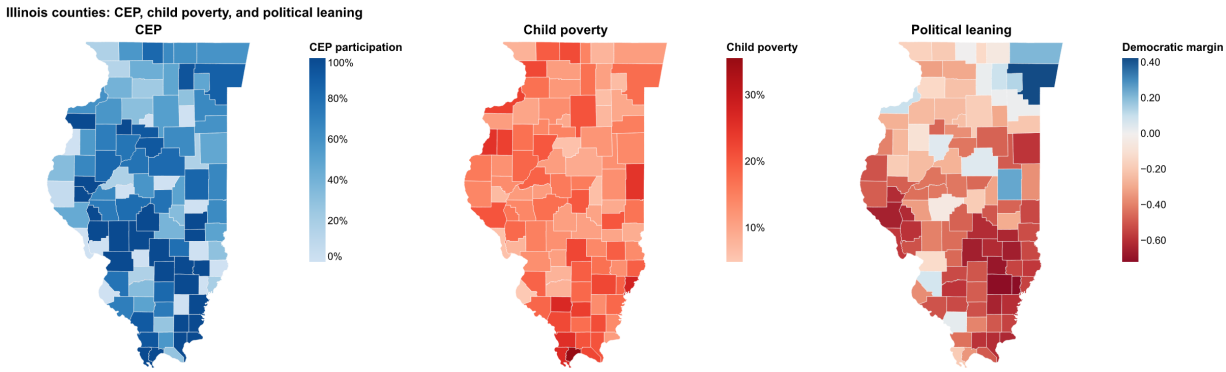
Visual Pattern

- CEP participation rate is positively correlated with the child food insufficiency rate.
- This means that the policy covers those in need to some extent.
- However, it is also clear that democratic-leaning states (blue states) tend to have a higher CEP participation rate.

These findings motivated us to a county-level analysis, taking IL as an example. In addtion, these spatial differences motivate formal statistical testing, which is whehter higher cep rates actually improves child-wellbeing (addressed in OLS part).

County level visualization

For the county-level analysis, we put the three data on the IL map, child poverty rate, cep participation rate, and democratic margin to see if there is any spatial pattern.



Visual Pattern

- It appears that cep participation rate is slightly positively correlated with the child poverty rate.
- This means that the policy covers those in need to some extent even at the county level.
- However, from the visual insepction, it is difficult to capture the spatial pattern or correlation between cep rate and the other two variables.
  Combined with the state-level visualization, it appears that the policy targeting is aligned with high-need regions. However, looking at the political effect, its correlation with the cep rate is pronounced, espetically at the state level.

4. Regression Analysis

This is discussed in the appendix

5. Streamlit

For dynamic visualization, we created the streamlit app where we can plot the various variables we discussed here. In addtion to what is dicussed in the plot, we added addtional plots (MO's data we discussed in the presentation and so on).
Noth that for county maps in the app, using altair kept showing errors, we decided to use matplotlib instead.

6. Discussion

What we learn

- The policy (cep participation rate) covers those in need in some degrees
- However, it may be associated with political affiliation at the state level

Limitations

- Only focus on IL data at the county level

7. Conclusion

Overall, the school meal program covers high-need regions. However, political leanings are also associated with the policy advancement, especially at the state level. Thus, institutional drive at the state level may be a key to further promote the policies.